Conformal-Mapping Summary

The Narrow-Channel Effect in MOSFET's with Semi-Recessed Oxide Structures

Li, Hong, Cheng & Chan, *IEEE Trans. Electron Devices*, vol. 37, no. 3, pp. 692-701, March 1990 — condensed companion to the full technical report (narrow_channel_semi_recessed_report.pdf) and slide deck (li1990_conformal_mapping_summary.pptx)

1. Method: Conformal Mapping of the Field-Oxide Region

A semi-recessed field-oxide MOSFET's narrow-channel behavior is governed by the 2-D electrostatics of the field-oxide cross-section next to the gate. Rather than solving $\nabla^2\phi=0$ directly on the awkward corner geometry, the paper maps the semi-infinite half-strip domain onto the upper half-plane via

$$\mathbf{u + iv = \cos[\pi(ix - y)/t_f]},$$

turning the problem into a classical Poisson-kernel / Schwarz-integral boundary-value problem, solvable as a Fourier series in n. The regime depends on whether the channel depletion depth t_d is below or above the recess depth t_1:

- **Case I** ($t_d < t_1$): depletion confined to the sidewall → inverse narrow-width effect ($\Delta V_{th} < 0$).
- **Case II** ($t_d > t_1$): depletion spills laterally under the field oxide (quarter-circle wing, radius $W_c=t_d-t_1$) → conventional effect ($\Delta V_{th} > 0$).
- Fully-recessed ($t_1=t_f-t_g$) and non-recessed ($t_1\to0$) structures fall out of the same formulas at limiting t_1 — no separate model needed.

The resulting series coefficients A,B (Case I) or A′,B′ (Case II) feed $V_{th}(W) = \phi_{ms} + 2\phi_f(1+a/W)/(1+b/W) + (Q'/C_{ox})/(1+b/W)$, with $a=2t_g(B-A)$, $b=-2t_gA$.

2. Verification: Beyond Dimensional Analysis

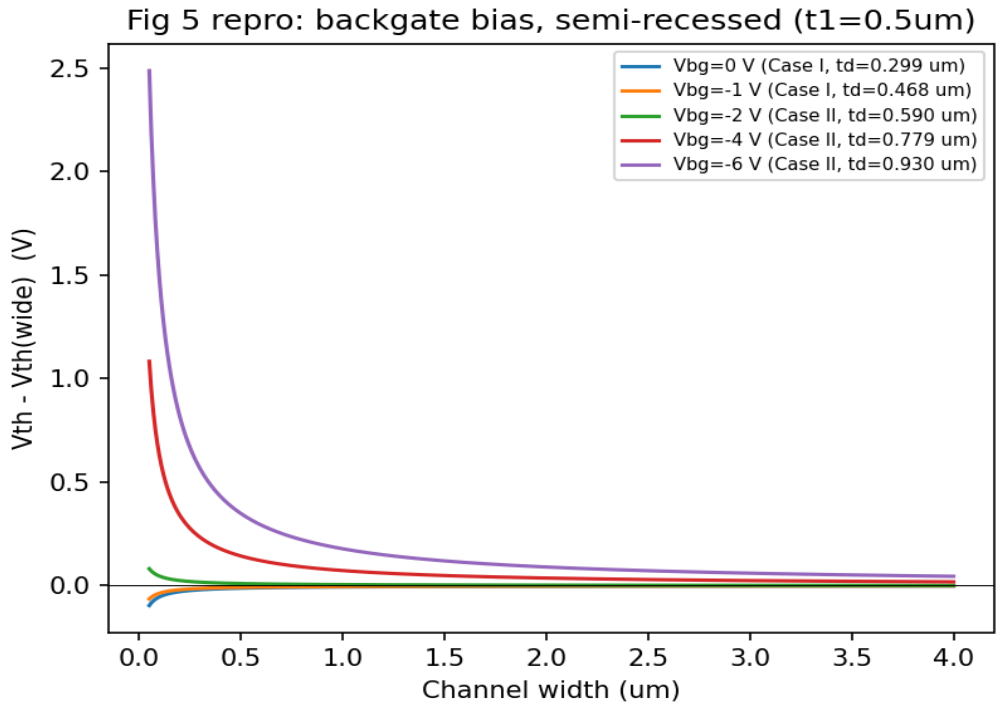
Dimensional analysis alone caught two real transcription issues in the paper's printed series (Eq. 11): a unit-inconsistent prefactor in A, and a missing ½ factor in $\sin(nkt_g)$. **Both fixes together were still not sufficient** — an independent finite-difference (FD) solve of the actual boundary-value problem (5-point Laplacian, sparse direct solve) disagreed with the corrected series, including in B, whose prefactor was never ambiguous.

The FD solver was validated two ways that do not depend on the paper's series at all:

- **Sharp-step analytic limit** ($t_d,t_g\to0$): the FD solution converges to a hand-derived closed form as the boundary condition simplifies to a step function (-0.4930 → -0.4659 vs. analytic -0.4652).
- **Case I/II boundary continuity** ($t_d\to t_1$): two independently-coded BVPs must agree in the shared limit — this check caught and fixed a real bug in the Case II lateral-wing flux integration bound.

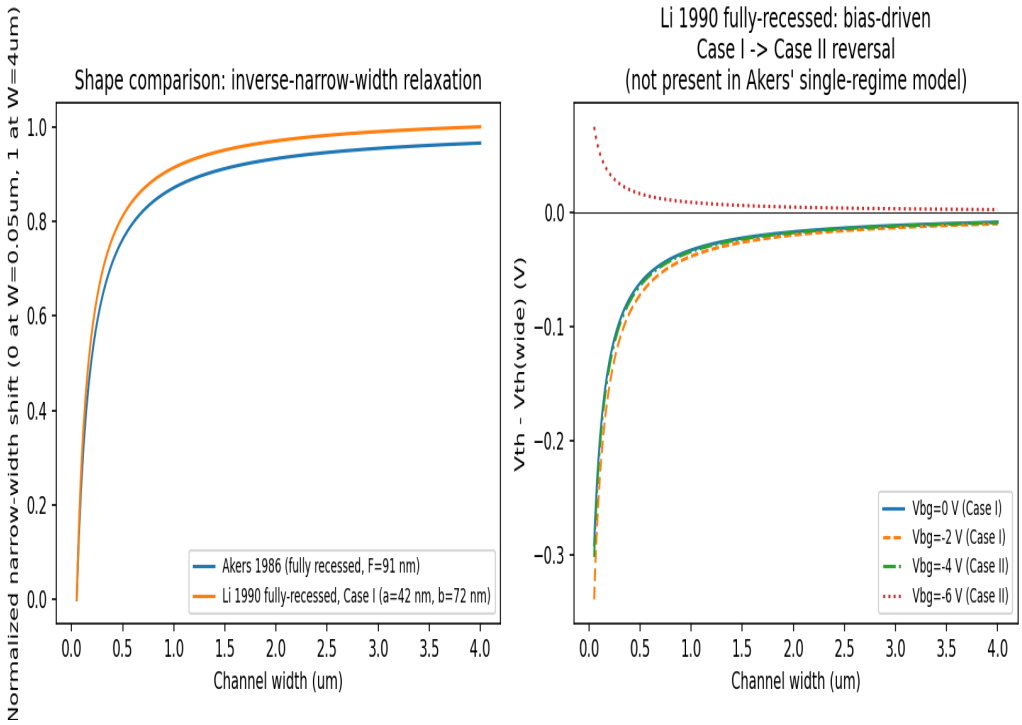
Given this, the FD solve — not the literal Eq. 11/17 series — is used as the verified ground truth throughout. The remaining series discrepancy's exact root cause is **reported as unresolved**, not forced to a clean fix.

3. Reproduced Result (Fig. 5)



Backgate-bias sweep, semi-recessed (t1=0.5 µm). Case I (Vbg=0,-1V) shows the inverse effect; Case II (Vbg=-2,-4,-6V) shows the conventional effect — matches the paper's own Case grouping exactly.

4. Cross-Check vs. Akers (1986)



Left: Li's fully-recessed Case I limit vs. Akers' single-regime model — same order-of-magnitude length scale (a=42.3 nm, b=71.8 nm vs. F=91 nm) from independent derivations. Right: Li's model additionally predicts a bias-driven Case I→Case II sign reversal that Akers' model cannot represent.

5. Bottom Line

- All 8 paper figures (Figs. 5-12) reproduced, matching qualitative trends and Case groupings.
- Fully-recessed length scales revised, post-FD-verification, from a=103 nm/b=124 nm (flawed series) to **a=42.3 nm, b=71.8 nm**.
- Full technical detail in narrow_channel_semi_recessed_report.pdf; slide-deck version in li1990_conformal_mapping_summary.pptx.